

Question number	Answer	Additional guidance	Mark
7(a)	A description that includes three of the following points: • {CFTR (protein) / (CFTR) protein} cannot function properly (1) • credit details of dysfunction (1) • mucus will be (very) {thick / sticky} (1) • gases will not be able to move through the mucus properly (1)	ACCEPT change in shape / structure / e.g. reduced transport of chloride ions out of the cell sodium ions move into the cell water leaves the mucus and enters the cell chloride ions remain in the cell ACCEPT oxygen (out of lungs) air (but not diffusion of air) mucus blocks airways	(3)

Question number	Answer	Additional guidance	Mark
7(b)(i)	• 13 / 13.0 / 12.97 (%) (1)		(1)

Question number	Answer	Additional guidance	Mark
7(b)(ii)	An explanation that includes three of the following points: • {more / improved} (genetic) screening (1) • more carriers identified (1) • because more {couples choosing IVF / (IVF) identification of affected embryos more reliable} (1) • more couple choosing not to have child (1) • as the frequency of the CF allele decreases in the population there will be a decrease in the CF phenotype (1) • more migration / change in population diversity (1)	ACCEPT named type of screening e.g. PGD ACCEPT fewer false negatives (in people) more (types of) mutations identified (in people) ACCEPT fewer false negatives (in embryo) more (types of) mutations identified (in embryo) fetus DO NOT ACCEPT baby / child ACCEPT more choosing adoption	(3)

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*7(c)	Regimen outlined • before inhalation, drug A must be administered • because they are looking at pre-treatment • a placebo needs to be given to another group • to see the effect of drug A over no pre-treatment / to eliminate bias • inhalation daily • as it is unlikely to stay in the body very long • run trial for weeks (minimum 4 weeks) • as results will not be immediate Design of trial • two / three groups of patients with CF (minimum 6 per group) • one to receive drug A and the other to receive a placebo • large number of patients in each group • for repeatability • patients should be of similar {age / sex / severity of CF} • for validity • should not be receiving other medications • as these may mask the effects of drug A and inhalation • double blind trial • to avoid bias What is measured • volume / mass of mucus removed • to see if clearance of mucus has improved • lung function tests / spirometer, peak flow • to see if lung function has improved • give a questionnaire • to assess improvements in quality of life • as these are the parameters being tested • use of statistics to analyse the results • to determine the significance of the data	(6)

			Additional guidance
Level 0	0	No awardable content	
Level 1	1-2	An explanation may be attempted but with limited interpretation or analysis of the scientific information and with a focus on mainly just one piece of scientific information. The explanation will contain basic information, with some attempt made to link knowledge and understanding to the given context.	Simple descriptions of investigation 1 mark = one description 2 marks = three descriptions
Level 2	3-4	An explanation will be given, with occasional evidence of analysis, interpretation and/or evaluation of both pieces of scientific information. The explanation shows some linkages and lines of scientific reasoning, with some structure.	Some explanation given of a trial 3 marks = one explanation 4 marks = two explanations
Level 3	5-6	An explanation is made that is supported throughout by sustained application of relevant evidence of analysis, interpretation and/or evaluation of both pieces of scientific information. The explanation shows a well-developed and sustained line of scientific reasoning, which is clear and logically structured.	More extensive explanation given 5 marks = three explanations 6 marks = four explanations for an investigation that will work